

Electromagnetic Waves

Displacement Current

Displacement current is that current which appears in a region in which the electric field (and hence electric flux) is changing with time.

Note: We have

$$I_D = \epsilon_0 \frac{d\phi_E}{dt} = \epsilon_0 \frac{d}{dt}(EA) = \epsilon_0 \frac{d}{dt} \left(\frac{qA}{\epsilon_0 A} \right) = \frac{dq}{dt} = I$$

[Charging of plate of a capacitor]

Modified Ampere's Circuital Law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(I + \epsilon_0 \frac{d\phi_E}{dt} \right)$$

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$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\phi_B}{dt} \text{ and } \oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\phi_E}{dt} \text{ (Maxwell's equations)}$$

These equations lead to the conclusion that, if either of the electric or magnetic field changes with time, the other field is induced in space. The net result of these interacting changing fields is the generation of electromagnetic disturbance, called electromagnetic waves which travel with the speed of light.

Mathematical Expression of EM Waves

$$E_y = E_0 \sin 2\pi \left(\frac{x}{\lambda} - \frac{t}{T} \right), B_z = B_0 \sin 2\pi \left(\frac{x}{\lambda} - \frac{t}{T} \right)$$

Important Characteristics of EM Waves

- EM waves are produced by accelerated charged particles.
- EM waves do not require any medium for their propagation. These waves can propagate in vacuum as well as in a medium.

Velocity of em waves in free space is given by

$$v = c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s}$$

Velocity of em waves in a medium is given by

$$v = \frac{c}{\sqrt{\mu_r k}}$$

$[\mu_r = \text{relative permeability of medium}]$

$[k = \text{dielectric constant of medium}]$

- EM waves are transverse in nature i.e. E and B are perpendicular to each other as well as perpendicular to the direction of propagation of the wave. E and B are related as follow:

$$\frac{E_0}{B_0} = c \text{ or } \frac{E}{B} = c$$

- EM waves carry energy, which is shared equally by electric and magnetic fields.

The average energy density of an EM wave is given by

$$u = u_E + u_B = 2u_E = 2u_B$$

$$\begin{aligned} \text{where } u_E &= \frac{1}{2} \epsilon_0 E^2 = \frac{1}{2} \epsilon_0 (Bc)^2 & \left[\because \frac{E}{B} = c \right] \\ &= \frac{1}{2} \epsilon_0 B^2 \left(\frac{1}{\sqrt{\mu_0 \epsilon_0}} \right)^2 = \frac{B^2}{2\mu_0} & \left[\because c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \right] \end{aligned}$$

- EM waves carry momentum and exert a radiation pressure

$$P = \frac{F}{A} = \frac{1}{A} \frac{dp}{dt} \text{ and momentum } p = \frac{U}{c}$$

- EM waves transport energy. The rate of energy of EM wave transport per unit area is represented by a quantity called

Poynting vector ($\vec{S}$) and is given by $\vec{S} = \frac{1}{\mu} (\vec{E} \times \vec{B})$

- Electric vector of an em wave is responsible for optical effects, as $E_0 \gg B_0$.

- Intensity of an EM wave is given by

$$I = \frac{1}{2} c \epsilon_0 E^2 = \frac{B^2 c}{2\mu_0}$$